SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VI – English Source-Aligned Cumulative Assembly Checkpoint

Six independently sealed bounded units; French lines 2139–2272

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Evidence boundary. This is a coherent reflowed assembly of the mathematical bodies of six independently source-reviewed and sealed working units. The corrected French arXiv TeX is the translation authority; the body occupies printed pp. 72–76, physical source-PDF pp. 65–68, and recomposed running pp. 57–60. The current jcreinhold `e7a259f` English Markdown is one comparison lineage only. Independent cumulative source and package review has passed. This bounded public checkpoint is not a critical edition, source certification, or authorization for any broader SGA claim.

Unit	Scope	French lines	Printed pp.	Physical pp.	Running pp.
1	Opening; 1.1–1.3	2139–2161	72	65	57
2	1.4–1.6	2163–2191	72–73	65–66	57–58
3	1.7–1.9	2193–2219	74	66–67	58–59
4	Proposition 2.1	2221–2227	75	67	59
5	2.2; Theorem 2.3 statement	2232–2257	75–76	67	59
6	Theorem 2.3 proof; bibliography	2259–2272	76	67–68	59–60

Exposé VI. The Functors $\operatorname{Ext}^\bullet_Z(X; F, G)$ and $\mathcal{E}xt^\bullet_Z(F, G)$

1. Generalities

1.1

Let $(X, \mathcal{O}_X)$ be a ringed space and let Z be a locally closed subset of X. Let F and G be $\mathcal{O}_X$-modules. We shall denote by $\operatorname{Ext}^i_Z(X; F, G)$ (respectively $\mathcal{E}xt^i_Z(F, G)$) the ith derived functor of

$$G \longmapsto \Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F, G)) \quad \text{(respectively } G \longmapsto \underline{\Gamma}_Z(\mathcal{H}om_{\mathcal{O}_X}(F, G))\text{)}.$$

Lemma 1.2. *The sheaf $\mathcal{E}xt^i_Z(F, G)$ is canonically isomorphic to the sheaf associated with the presheaf*

$$U \longmapsto \operatorname{Ext}^i_{Z\cap U}(U; F|_U, G|_U).$$

This follows from ([Tôhoku], 3.7.2) and from the fact that

$$\Gamma(U; \underline{\Gamma}_Z(\mathcal{H}om_{\mathcal{O}_X}(F, G))) \simeq \Gamma_{Z\cap U}\Big(\mathcal{H}om_{\mathcal{O}_X|_U}(F|_U, G|_U)\Big)$$

canonically.

Theorem 1.3 (Excision theorem). *Let V be an open subset of X containing Z. Then there is an isomorphism of cohomological functors*

$$\operatorname{Ext}^\bullet_X(X; F, G) \simeq \operatorname{Ext}^\bullet_V(V; F|_V, G|_V). \tag{1.3.1}$$

Indeed, if $G^\bullet$ is an injective resolution of G, then $G^\bullet|_V$ is an injective resolution of $G|_V$. The theorem follows immediately.

1.4

Let $\mathcal{O}_{X,Z}$ be the $\mathcal{O}_X$-module defined by the following conditions ([Godement], 2.9.2):

$$\mathcal{O}_{X,Z}|_{X-Z} = 0 \qquad \text{and} \qquad \mathcal{O}_{X,Z}|_Z = \mathcal{O}_X|_Z.$$

We have seen that, for every $\mathcal{O}_X$-module H, there is a functorial isomorphism

$$\Gamma_Z(H) \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, H).$$

Consequently, there are isomorphisms functorial in F and G:

$$\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G)) \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, \mathcal{H}om_{\mathcal{O}_X}(F,G)), \tag{1.4.1}$$
$$\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G)) \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z} \otimes_{\mathcal{O}_X} F, G), \tag{1.4.2}$$
$$\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G)) \simeq \mathrm{Hom}_{\mathcal{O}_X}(F, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, G)) = \mathrm{Hom}_{\mathcal{O}_X}(F, \underline{\Gamma}_Z(G)). \tag{1.4.3}$$

In particular, it follows from (1.4.2) that there is a ∂-functorial isomorphism in F and G,

$$\theta\colon\ \mathrm{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_{X,Z} \otimes_{\mathcal{O}_X} F, G) \xrightarrow{\ \sim\ } \mathrm{Ext}^i_Z(X; F, G).$$

1.5

By definition, the functor $G \longmapsto \Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G))$ is the composite of $G \longmapsto \mathcal{H}om_{\mathcal{O}_X}(F,G)$ and Γ_Z. Since Γ_Z is left exact (I 1.9), since $\mathcal{H}om_{\mathcal{O}_X}(F,G)$ is flasque when G is injective, and since Γ_Z is exact on flasque sheaves (I 2.12), it follows from ([Tôhoku], 2.4.1) that there is a spectral functor abutting to $\mathrm{Ext}^\bullet_Z(X;F,G)$ whose initial term is

$$H^p_Z(X, \mathcal{E}xt^q_{\mathcal{O}_X}(F,G)).$$

On the other hand, it follows from (1.4.3) that $\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G))$ is the composite of $G \longmapsto \underline{\Gamma}_Z(G)$ and $H \longmapsto \mathrm{Hom}_{\mathcal{O}_X}(F,H)$. Since $\underline{\Gamma}_Z$ takes injectives to injectives (I 1.4), it follows from ([Tôhoku], 2.4.1) that there is a spectral functor abutting to $\mathrm{Ext}^\bullet_Z(X;F,G)$ whose initial term is

$$\mathrm{Ext}^p_{\mathcal{O}_X}(X; F, \mathcal{H}^q_Z(G)).$$

Finally, it follows from (1.4.2) and the Ext spectral sequence that there is a spectral functor abutting to $\mathrm{Ext}^\bullet_Z(X;F,G)$ whose initial term is $H^p(X; \mathcal{E}xt^\bullet_Z(F,G))$. Hence:

Theorem 1.6. *There are three spectral functors abutting to* $\mathrm{Ext}^\bullet_Z(X;F,G)$ *whose initial terms are respectively*

$$H^p_Z(X, \mathcal{E}xt^q_{\mathcal{O}_X}(F,G)), \tag{1.6.1}$$
$$H^p(X, \mathcal{E}xt^q_Z(F,G)), \tag{1.6.2}$$
$$\mathrm{Ext}^p_{\mathcal{O}_X}(X; F, \mathcal{H}^q_Z(G)). \tag{1.6.3}$$

1.7

Let Z' be a closed subset of Z, and set $Z'' = Z - Z'$. There is an exact sequence

$$0 \longrightarrow \mathcal{O}_{X,Z''} \longrightarrow \mathcal{O}_{X,Z} \longrightarrow \mathcal{O}_{X,Z'} \longrightarrow 0. \tag{1.7.1}$$

It generalizes the exact sequence of ([Godement], 2.9.3). This exact sequence splits locally; hence, for every $\mathcal{O}_X$-module F, there is another exact sequence

$$0 \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X,Z''} \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X,Z} \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X,Z'} \longrightarrow 0. \tag{1.7.2}$$

Now let G be an $\mathcal{O}_X$-module. Applying the functor $\mathrm{Hom}_{\mathcal{O}_X}(\bullet, G)$ to (1.7.2), one obtains from (1.4.2) and the long exact sequence of Ext the following theorem.

Theorem 1.8. *Let Z be a locally closed subset of X, let Z' be a closed subset of Z, and set $Z'' = Z - Z'$. There is an exact sequence, functorial in F and G,*

$$0 \longrightarrow \mathrm{Hom}_{Z'}(F, G) \longrightarrow \mathrm{Hom}_Z(F, G) \longrightarrow \mathrm{Hom}_{Z''}(F, G) \longrightarrow \mathrm{Ext}^1_{Z'}(F, G) \longrightarrow \cdots$$
$$\cdots \longrightarrow \mathrm{Ext}^i_Z(F, G) \longrightarrow \mathrm{Ext}^i_{Z''}(F, G) \longrightarrow \mathrm{Ext}^{i+1}_{Z'}(F, G) \longrightarrow \cdots .$$

Corollary 1.9. *Let Y be a closed subset of X, and set $U = X - Y$. There is an exact sequence, functorial in F and G,*

$$0 \longrightarrow \mathrm{Hom}_Y(F, G) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(F, G) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X|U}(F|_U, G|_U) \longrightarrow \mathrm{Ext}^1_Y(F, G) \longrightarrow \cdots$$
$$\cdots \longrightarrow \mathrm{Ext}^i_{\mathcal{O}_X}(F, G) \longrightarrow \mathrm{Ext}^i_{\mathcal{O}_X|U}(F|_U, G|_U) \longrightarrow \mathrm{Ext}^{i+1}_Y(F, G) \longrightarrow \cdots .$$

This corollary is an immediate consequence of Theorems (1.3) and (1.8).

2. Applications to Quasi-Coherent Sheaves on Preschemes

Proposition 2.1. *Let X be a locally Noetherian prescheme. For every locally closed subset Z of X, every coherent $\mathcal{O}_X$-module F, and every quasi-coherent $\mathcal{O}_X$-module G, the sheaves $\mathcal{E}xt^i_Z(F, G)$ are quasi-coherent.*

As in (1.6.3), one shows that the sheaves $\mathcal{E}xt^i_Z(F, G)$ are obtained as the abutment of a spectral sequence whose initial term is

$$\mathcal{E}xt^p_{\mathcal{O}_X}(F, \mathcal{H}^q_Z(G)).$$

By (II, Corollary 3), the $\mathcal{H}^q_Z(G)$ are quasi-coherent, and therefore so are the $\mathcal{E}xt^p_{\mathcal{O}_X}(F, \mathcal{H}^q_Z(G))$, since F is coherent. The proposition follows immediately.

2.2

Let Y now be a closed subprescheme of X, and let $\mathcal{I}$ be an ideal defining Y. Let m and n be integers such that $m \geq n \geq 0$. We denote by $i_{n,m}$ the canonical map

$$\mathcal{O}_{Y_m} = \mathcal{O}_X/\mathcal{I}^{m+1} \longrightarrow \mathcal{O}_X/\mathcal{I}^{n+1} = \mathcal{O}_{Y_n}$$

and by j_n the map $\mathcal{O}_{X,Y} \to \mathcal{O}_{Y_n}$. The $(\mathcal{O}_{Y_n}, i_{n,m})$ form a projective system, and the j_n are compatible with the $i_{n,m}$.

Applying the functor $\mathrm{Ext}^i_{\mathcal{O}_X}(F \otimes \bullet, G)$, we obtain a morphism

$$\varphi' \colon \varinjlim_n \mathrm{Ext}^i_{\mathcal{O}_X}(X; F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \mathrm{Ext}^i_{\mathcal{O}_X}(X; F \otimes \mathcal{O}_{X,Y}, G);$$

this is a morphism of cohomological functors in G. The morphism

$$\varphi \colon \varinjlim_n \mathrm{Ext}^i_{\mathcal{O}_X}(X; F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \mathrm{Ext}^i_Y(X; F, G),$$

which is the composite $\theta \circ \varphi'$ (cf. (1.4)), is therefore also a morphism of cohomological functors in G.

In the same way one defines

$$\underline{\varphi} \colon \varinjlim_n \mathcal{E}xt^i_{\mathcal{O}_X}(F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \mathcal{E}xt^i_Y(X; F, G).$$

Theorem 2.3. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X defined by a coherent ideal $\mathcal{I}$, let F be a coherent $\mathcal{O}_X$-module, and let G be a quasi-coherent $\mathcal{O}_X$-module. Then:*

a) $\underline{\varphi}$ *is an isomorphism.*

b) If X is Noetherian, φ *is an isomorphism.*

Proof of Theorem 2.3

The proof of b) is almost word for word the same as that of (II 6 b)), using the spectral sequence (1.6.2), so we do not reproduce it.

For the proof of a), by (2.1) we may assume that X is affine with ring A, that F (respectively G) is defined by an A-module M (respectively N), and that $\mathcal{I}$ is defined by an ideal I. It is enough to prove that the homomorphism

$$\varinjlim_{n} \mathrm{Ext}^i_A(M/I^n M, N) \longrightarrow \mathrm{Ext}^i_Y(X, F, G) \tag{2.3.1}$$

induced by $\underline{\varphi}$ is an isomorphism.

Indeed, for $i = 0$, the two sides of (2.3.1) may be canonically identified with the submodule of $\mathrm{Hom}_A(M, N)$ consisting of the elements of $\mathrm{Hom}_A(M, N)$ annihilated by a power of I. One then sees that the homomorphism (2.3.1) is simply the identity map.

The functor

$$N \longmapsto \varinjlim_{n} \mathrm{Ext}^{\bullet}_A(M/I^n M, N)$$

is a universal ∂-functor. We shall show that the same holds for the functor $N \longmapsto \mathrm{Ext}^{\bullet}_Y(M, N)$. Indeed, if N is an injective module, then by (II, 9 and 11), $\mathcal{H}^q_Y(N) = 0$ for $q \neq 0$; and by (IV.2.2), $H^0_Y(N)$ is injective.

It follows that

$$\mathrm{Ext}^p_{\mathcal{O}_X}(X; M, \mathcal{H}^q_Y(N)) = 0 \qquad \text{if } p + q \neq 0.$$

Thus, by (1.6.3), $\mathrm{Ext}^i_Y(M, N) = 0$ for $i \neq 0$ when N is injective. This completes the proof.

Bibliography

The same references as those listed at the end of Exposé I, cited respectively as [Tôhoku] and [Godement].